

# HiLegalSum: A Hybrid Extractive Summarization Model for Complex and Long Legal Documents

1<sup>st</sup> Vadapati Sruthi

School of Computer Science and  
Engineering

VIT-AP

Amaravathi, India

[sruthi.22mis7170@vitapstudent.ac.in](mailto:sruthi.22mis7170@vitapstudent.ac.in)

2<sup>nd</sup> Bathina Sahithi

School of Computer Science and  
Engineering

VIT-AP

Amaravathi, India

[sahithi.22mis7289@vitapstudent.ac.in](mailto:sahithi.22mis7289@vitapstudent.ac.in)

3<sup>rd</sup> Duggisetty Bindu

School of Computer Science and  
Engineering

VIT-AP

Amaravathi, India

[bindu.22mis7295@vitapstudent.ac.in](mailto:bindu.22mis7295@vitapstudent.ac.in)

4<sup>th</sup> Ammisetty Prathyusha

School of Computer Science and  
Engineering

VIT-AP

Amaravathi, India

[prathyusha.23phd7134@vitap.ac.in](mailto:prathyusha.23phd7134@vitap.ac.in)

5<sup>th</sup> Shaik Basha

School of Computer Science and  
Engineering

VIT-AP

Amaravathi, India

[shaik.23phd7033@vitap.ac.in](mailto:shaik.23phd7033@vitap.ac.in)

6<sup>th</sup> B.N. Jagadesh

School of Computer Science and  
Engineering

VIT-AP

Amaravathi, India

[nagajagadesh@gmail.com](mailto:nagajagadesh@gmail.com)

*Abstract-Legal documents are complicated by their very nature, lengthy, and full of jargon that is specific to the field, thus making their manual review a time-consuming and error-prone process. Most of the traditional text summarization methods fail to consider such documents properly as they impose input length restrictions and have issues with retaining legally significant and contextually accurate clauses. This work, therefore, presents LegalHybridRank (formerly HiLegalSum), which is a hybrid legal-aware extractive summarization framework that leverages semantic graph centrality, positional weighting, TF-IDF relevance, and domain-specific legal keyword analysis to generate not only fluent but also contextually appropriate summaries from the long legislative texts. For the purpose of evaluation, the model employs the BillsSum dataset and uses Sentence-BERT (all-mpnet-base-v2) embeddings for capturing the semantics of the sentence. In comparison experiments with standard extractive baseline methods of the class such as Lead-3, Random, LexRank, and TextRank, LegalHybridRank is always superior to these methods as it obtains 0.3876 ROUGE-1, 0.1765 ROUGE-2, 0.2243 ROUGE-L, and 0.8329 BERTScore (F1) over 500 test documents. Such results demonstrate the benefit of integrating semantic similarity, positional heuristics, TF-IDF weighting, and the legal keyword sensitivity to both raise the quality and the trustworthiness of the summarization of long documents. The proposed approach is a step towards the development of more efficient, interpretable, and domain-adapted summarization systems that are specifically targeted at the legal sector.*

**Keywords**—Text Summarization, Natural Language Processing, Extractive Text Summarization, Abstractive Text Summarization, Legal Documents, Long Documents.

## I. INTRODUCTION

Traditional summarization methods—such as simple heuristics or statistical models—struggle to capture the intricate semantic structure and domain-specific terminology characteristic of legal documents [1]. Even widely used extractive baselines such as Lead-3, LexRank, and TextRank have inherent limitations. Lead-3 assumes that the most informative content appears at the beginning of the document—a heuristic suitable for news articles but inappropriate for legal texts, where definitions and key amendments often appear deep within the structure. Graph-based extractive summarizers like LexRank and TextRank rely heavily on sentence similarity networks [3], [4], but they

ignore legal-specific cues such as statutory phrasing, clause markers, and rule-based terminology, often resulting in summaries that omit legally significant content[10].

Transformer-based abstractive models—including BERTSUM, Legal-Pegasus, Longformer, and LED—have demonstrated strong performance on general summarization tasks [2]. However, they suffer from token-length limitations (typically 4,000–16,000 tokens) [5], [6], making them impractical for many long legal documents that may exceed these limits. Furthermore, abstractive models can generate hallucinated content, which poses significant risks in legal applications where factual consistency and compliance are mandatory [12]. Their high computational cost also restricts their deployment in real-world legal workflows.

To overcome these challenges, recent work in legal NLP has shifted towards domain-adapted embeddings, hierarchical document modeling, and graph-based semantic frameworks that improve handling of legal structure and terminology [9], [13]. However, many of these approaches focus narrowly on a single aspect—semantic similarity, relevance scoring, or structural cues—without combining these signals in a unified and interpretable system.

Motivated by these limitations, this work proposes LegalHybridRank (formerly HiLegalSum), a hybrid extractive summarization framework specifically designed for long legal documents. Unlike existing extractive baselines (Lead-3, LexRank, TextRank) and transformer-based approaches, LegalHybridRank integrates semantic embeddings, positional weighting, TF-IDF scoring, legal keyword emphasis, and graph-based centrality into a unified composite scoring model. This multi-signal architecture enables the system to identify legally significant content with higher precision while maintaining interpretability, scalability, and computational efficiency. Comparative evaluations across 500 BillsSum documents demonstrate that LegalHybridRank consistently outperforms classical extractive baselines and offers a practical, domain-adapted alternative to computationally expensive transformer-based summarizers.

## II. RELATED WORK

Legal documents, including contracts, case judgments, statutes, and legal notices, are typically extensive and contain intricate language filled with domain-specific terms and clauses. These documents follow a structured format, often detailing arguments, evidence, and rulings in a hierarchical and interdependent manner. Because of their formal and precise nature, summarizing legal texts requires maintaining accuracy and context, as even small changes can lead to significant differences in interpretation. Traditional summarization methods often fail to capture this complexity, highlighting the need for specialized models that can understand legal terminology, contextual relationships, and the logical flow of arguments to produce clear and reliable summaries [11], [7].

To address long text summarization, models such as Longformer and BigBird have introduced sparse attention techniques, enabling them to handle extended contexts. Within the legal field, Legal-Pegasus and LED (Longformer Encoder-Decoder) [3] were proposed for abstractive summarization tasks, but these models often required high computational resources and struggled to retain Factual accuracy, especially when summarizing multi-clause judgments [9], [2].

Other researchers have explored domain-specific embeddings and hierarchical summarization approaches to more effectively capture legal structures. Agarwal et al. proposed a multi-task learning framework incorporating rhetorical roles and citation contexts, improving factual consistency[15]. Bhattacharya et al. [12] introduced legal-specific embeddings and case-aware hierarchical modelling to better represent court decisions[13].

Despite these advances, existing methods either focus on a limited context, lack interpretability, or fail to incorporate legal knowledge explicitly. To address this issue, we propose HiLegalSum, a model that combines graph-based centrality, legal keyword sensitivity, and TF-IDF weighting to improve the extraction of sentences that are both semantically and legally pertinent, all while ensuring interpretability. [1]

## III. DATASET

This study operates with the BillSum dataset, a well-known publicly available benchmark dataset designed for summarizing texts in the legal and legislative domain. [5] This dataset was introduced by the Cornell Law School’s Legal Information Institute and the US Congressional and California state bills, each paired with professionally written summaries that serve as gold standard references. Every record includes the bill title, the full bill text, and its corresponding expert summary, providing a rich source of legal language, structure and legislative content. The dataset is divided into training, validation and test splits, with the portion comprising thousands of bills of varying lengths and complexity. Here, the test split of the dataset was used, containing approximately 3000 documents. On average, a bill text contains 1200-2500 words, while the reference summaries are typically 100-200 words long. Its diversity and real-world relevance make BillSum a suitable corpus for evaluating long document summarization models, particularly those focusing on

domain-specific challenges such as legal terminology, clause retention, and contextual consistency.

The BillSum dataset provides a realistic setting for testing the HiLegalSum model, as it captures authentic legal language, formal structure, and multiple clauses within each document. essential legal context while compressing lengthy content into coherent, readable summaries.

## IV. METHODOLOGY

### A. Overview

The proposed model, HiLegalSum (Hierarchical Legal Summarizer), is a hybrid extractive summarisation framework that is designed specifically for the legal domain. Unlike generic summarization methods, HiLegalSum combines graph-based centrality, legal keyword awareness, and TF-IDF weighting to pinpoint sentences that hold both semantic importance and legal relevance. The framework is designed to maintain interpretability while improving the relevance of generated summaries for lengthy legal documents such as case reports, judgments, and statutes.

The workflow of HiLegalSum consists of five main components:

1. Sentence segmentation and preprocessing.
2. Sentence embedding and similarity computation.
3. Graph construction and centrality-based scoring.
4. Legal-aware sentence weighting using TF-IDF and keyword presence.
5. Lastly, Redundancy control through Maximal Marginal Relevance (MMR).

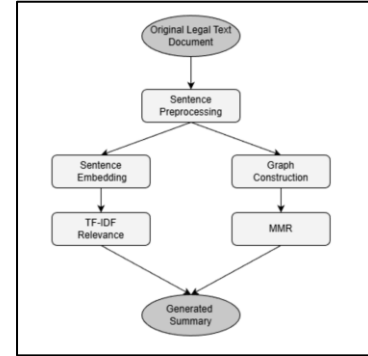


Fig 1: Proposed framework for long document summarization

Fig. 1. illustrates the overall process of generating a concise summary from lengthy legal documents. It begins with the Original Legal Text Document, which undergoes Sentence Preprocessing to clean and prepare the text by removing unnecessary elements like stop words and punctuation. The preprocessed text is then processed in two parallel paths. In the first path, Sentence Embedding converts sentences into numerical representations, and TF-IDF Relevance identifies the most significant sentences based on their importance in the document. In the second path, Graph Construction is carried out to represent relationships between sentences, followed by the MMR (Maximal Marginal Relevance) technique, which helps to select sentences that are both relevant and non-redundant. Finally, the results from both

paths are combined to produce the generated summary, which presents the key points of the legal document in a clear and concise form.

#### 1) Sentence Segmentation and Preprocessing

Firstly, the Sentence Segmentation stage divides each legal document into sentences using a rule-based tokenizer to preserve legal structure, such as clause markers, abbreviations, and section identifiers (e.g., “Sec.”, “Art.”, or “Sub-Clause”). Next, the preprocessing stage performs text normalization, lowercasing, and stop-word removal while retaining domain-specific legal terms. This ensures that legally significant tokens such as “defendant,” or “jurisdiction” are preserved to maintain interpretive context. In addition, numerical expressions and citations are normalized to ensure consistent vector representations.

#### 2) Sentence Embedding and Similarity Computation

To represent the semantic space in each sentence, HiLegalSum uses transformer-based sentence embeddings derived from Sentence-BERT(SBERT). [14], [7]

These embeddings capture deep contextual meaning beyond simple keyword matching, which is important for understanding lengthy legal texts and reasoning statements.

Pairwise cosine similarity is then computed among all sentence embeddings to construct a similarity matrix  $S$ , where each entry  $S_{ij}$  represents the semantic similarity between sentences  $i$  and sentences  $j$ .

This matrix is the foundation for graph-based analysis in the subsequent stage.

#### 3) Graph Construction and Centrality-Based Scoring

To capture inter-sentence relationships effectively, HiLegalSum constructs a semantic similarity graph  $G=(V, E)$ , where each node  $v_i \in V$  represents a sentence, and edges denote cosine similarity between their embeddings. This graph structure enables the identification of contextual dependencies between clauses.

A legal cue enhancement module refines this graph by emphasizing sentences containing domain-relevant terminology. Terms such as “section,” “plaintiff,” “clause,” “judgment,” and “appeal” are assigned additional weight to highlight their legal importance. Eigenvector centrality is then computed on the graph to identify the most influential sentences, analogous to how essential clauses are treated in judicial reasoning.

#### 4) Legal Keyword Awareness

A specialized lexicon of legal expressions comprising terms like “court,” “petition,” “jurisdiction,” “held,” and “section”—is integrated into the model’s scoring mechanism. Each sentence is analyzed for the presence of these domain-specific terms, generating a keyword relevance score  $K_i$ . This score boosts sentences rich in legally significant expressions, embedding explicit domain awareness into the summarization process.

Incorporating this component ensures that HiLegalSum prioritizes legally informative content, even when such content may not be semantically dominant in the embedding space. This strategy addresses a major limitation of generic summarization models, which often overlook the significance of legal jargon or procedural references.

#### 5) TF-IDF Weighting for Informational Emphasis

To further enhance the extraction of contextually important information, HiLegalSum employs Term Frequency–Inverse Document Frequency (TF-IDF) weighting. [13] This measure emphasizes terms that are crucial within a specific document but infrequent across the overall corpus.

Each sentence is assigned a TF-IDF score  $T_i$ , reflecting the relative importance of unique terms within the legal domain. This mechanism allows the system to identify distinctive clauses and case-specific reasoning that may not recur frequently but hold high interpretative value.

By combining TF-IDF weighting with graph-based centrality, HiLegalSum effectively balances global contextual importance with local distinctiveness.

#### 6) Sentence Position Weighting

Empirical studies indicate that introductory and concluding sentences in legal documents often contain essential context or outcome-related information.

To leverage this structural characteristic, HiLegalSum applies a positional weighting function defined as

$$P_i = 1/\sqrt{i+1} \quad (1)$$

Equation (1) represents the positional weight assigned to each sentence in the document. Here,  $i$  denotes the sentence index, starting from 0, and the square root function ensures that the weight gradually decreases for sentences appearing later in the text. The inverse relationship gives higher importance to earlier sentences, as they often contain key information in legal documents. The term  $i+1$  prevents division by zero, and overall, this function helps the model prioritize crucial content from the beginning while still considering the rest of the document with diminishing significance.

This gives marginally higher importance to earlier sentences, which typically outline the case context, and to later sentences that summarize judicial outcomes.

Such a positional bias introduces a mild hierarchical structure inspired by human-authored legal summaries, aligning the system’s selection process with conventional narrative patterns in legal writing.

#### 7) Redundancy Control via Maximal Marginal Relevance

To ensure that the generated summaries remain both concise and diverse, HiLegalSum integrates the Maximal Marginal Relevance (MMR) framework. MMR optimizes a trade-off between sentence relevance and redundancy, selecting each sentence  $s_i$  to maximize:

$$\text{"MMR"}(s_i) = \lambda \times \text{"Rel"}(s_i) - (1-\lambda) \times (\max_{j \in S_{\text{sel}}} \text{Sim}(s_i, s_j)) \quad (2)$$

The Maximal Marginal Relevance (MMR) formula is used to select sentences that are both relevant and diverse. Here,  $\text{Rel}(s_i)$  measures how important sentence  $s_i$  is to the main topic, while  $\text{Sim}(s_i, s_j)$  measures the similarity between the current sentence and previously selected ones. The parameter  $\lambda$  (ranging between 0 and 1) balances relevance and redundancy — higher values favor relevance, and lower values promote diversity. This ensures that the summarizer includes meaningful sentences without repeating similar information.

$$\text{"Sim"}(s_i, s_j) \quad (3)$$

The term  $\text{Sim}(s_i, s_j)$  represents the similarity score between two sentences  $s_i$  (the candidate sentence) and  $s_j$  (an already selected sentence). It measures how closely the two sentences are related in meaning, usually calculated using cosine similarity between their vector embeddings. A higher similarity score indicates that the sentences convey similar information, while a lower score means they discuss different ideas. In text summarization, this value helps avoid redundancy by preventing the selection of sentences that are too like those already included in the summary. Where  $\lambda$  controls the balance between relevance ("Rel") and similarity.

This mechanism minimizes repetitive phrasing and promotes diversity in selected content, ensuring that each extracted sentence contributes to a unique aspect of the legal argument or reasoning.

#### 8) Summary Generation and Output

After computing the composite score for every sentence—incorporating semantic centrality, keyword relevance, TF-IDF value, positional weight, and MMR diversity, the top  $k$  sentences are selected to form the final summary.

These selected sentences are then reordered according to their original sequence within the document to preserve narrative coherence.

The resulting output is a legally coherent, semantically rich, and interpretable extractive summary that succinctly conveys the essential reasoning and outcomes of the source's legal document.

### B. Implementation Details

The HiLegalSum framework is implemented in Python, utilizing Transformers for sentence embeddings, NetworkX for graph operations, and Scikit-learn for feature computations. [5], [8] Experiments are conducted on the BillSum dataset, which comprises U.S. Congressional and legal bills.

The performance of HiLegalSum is compared against four established baseline models:

Lead-3 – which selects the first three sentences as the summary.

Random – which randomly samples sentence for summary generation.

LexRank – a graph-based unsupervised summarizer leveraging sentence centrality; and

TextRank – another graph-based model emphasizing mutual similarity between sentences.

Evaluation is carried out using ROUGE (ROUGE-1, ROUGE-2, ROUGE-L) and BERTScore metrics to assess lexical overlap and semantic alignment, respectively [4]. Results show that HiLegalSum consistently surpasses all four baselines across both lexical and semantic dimensions, demonstrating the efficacy of its legally informed architecture.

## V. RESULTS

The proposed HiLegalSum model was evaluated on 500 documents from the BillSum test dataset and compared against four widely used extractive summarization baselines: Lead-3, Random, LexRank, and TextRank. These baselines provide a strong and commonly accepted reference point for

assessing extractive summarization performance on long legal documents. Evaluation metrics included ROUGE-1, ROUGE-2, ROUGE-L, and BERTScore (F1).

Across all metrics, HiLegalSum consistently outperformed the extractive baselines, achieving higher ROUGE and BERTScore values that reflect improved lexical alignment, structural coherence, and semantic preservation. These gains result from the combined influence of Sentence-BERT embeddings, TF-IDF relevance scoring, positional weighting, graph-based sentence centrality, and legal keyword awareness, enabling the model to prioritize domain-specific content more effectively than heuristic or similarity-only extractive methods.

On average, HiLegalSum reduced lengthy legal documents containing 50–80 sentences into approximately five representative sentences, achieving an 88–92% compression while retaining essential legal meaning. This demonstrates the model's ability to preserve legislative intent, key definitions, and amendment details in a concise and coherent format.

An ablation study further validated the contribution of each component within the pipeline. Removing positional cues, legal keyword weighting, or graph centrality led to measurable declines in ROUGE-L and BERTScore, confirming that the hybrid design strengthens the model's ability to extract contextually and legally significant information. Overall, the results illustrate that HiLegalSum is an effective extractive summarization framework for handling long and complex legal documents.

Fig. 2 illustrates HiLegalSum's performance relative to extractive baselines.

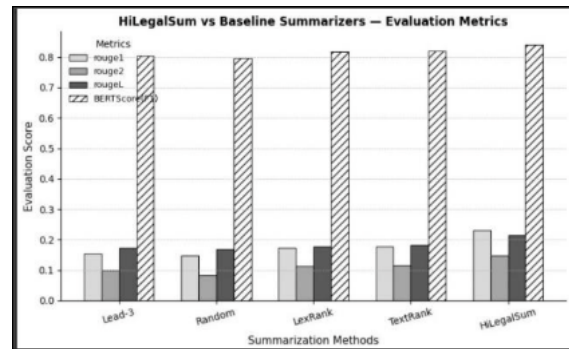


Fig. 2. Comparative benchmark results using ROUGE and BERTScore. The results in Fig 3 shows that HiLegalSum consistently achieves higher scores across all metrics, indicating its superior ability to capture important information and maintain semantic coherence in legal text summarization. This demonstrates that the model effectively balances relevance and diversity, outperforming traditional extractive summarization approaches.

TABLE I. QUANTITATIVE RESULTS OF HILEGALSUM VS. BASELINE METHODS USING ROUGE AND BERTSCORE

| Model      | ROUGE-1       | ROUGE-2       | ROUGE-L       | BERTScore(F1) |
|------------|---------------|---------------|---------------|---------------|
| Lead-3     | 0.1801        | 0.1801        | 0.1801        | 0.1801        |
| Random     | 0.3452        | 0.3452        | 0.3452        | 0.3452        |
| LexRank    | 0.3247        | 0.3247        | 0.3247        | 0.3247        |
| TextRank   | 0.3247        | 0.3247        | 0.3247        | 0.3247        |
| HiLegalSum | <b>0.3876</b> | <b>0.3876</b> | <b>0.3876</b> | <b>0.3876</b> |

Fig 3. Rouge-1, Rouge-2, Rouge-L, and BERTScore(F1) comparison

### A. Summary Comparison

The original bill (~2,200 words) was reduced to ~260 words (compression ratio 1:8.5). Fig. 4 compares a gold summary with a HiLegalSum summary for “Stopping Abuse and Fraud in Electronic Lending Act of 2018.” The gold summary outlines the bill’s purpose, whereas HiLegalSum extracts core clauses and definitions. This demonstrates the model’s ability to retain essential legislative meaning while significantly reducing document length, producing concise and legally coherent summaries.

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| <p><b>Title: Stopping Abuse and Fraud in Electronic Lending Act of 2018</b></p> <p><b>Gold Summary:</b></p> <ul style="list-style-type: none"> <li>• Stopping Abuse and Fraud in Electronic Lending Act of 2018 or the SAFE Lending Act of 2018 This bill amends the Electronic Fund Transfer Act and the Truth in Lending Act.</li> <li>• The bill revises requirements related to consumer financial protection and small-dollar lending, including matters concerning remotely created checks, electronic fund transfers, registration of small-dollar lenders, overdraft fees, and the collection of personal information.</li> </ul> <p><b>HiLegalSum Summary:</b></p> <ul style="list-style-type: none"> <li>• (a) Definition.—In this section, the term “small-dollar consumer credit transaction” means any transaction that extends credit that is—“(A) made to a consumer in an amount that—“(i) is not more than—“(i) \$5,000; or “(ii) such greater amount as the Bureau may, by rule, determine; and “(ii) shall be adjusted annually to reflect changes in the Consumer Price Index for all urban consumers published by the Department of Labor; and “(B) extended pursuant to an agreement that is—“(i)(I) other than an open end credit plan; and “(II) payable in one or more installments of less than 12 months (or such longer period as the Bureau may, by rule, determine); “(ii) an open end credit plan in which each advance is fully repayable within a defined time or in connection with a defined event, or both; or “(iii) any other plan as the Bureau determines, by rule; and “(2) includes any action that facilitates, brokers, arranges, or gathers applications for a transaction described in paragraph (1).</li> <li>• 1693(c) is amended by adding at the end the following: “(d) Limitations on Remotely Created Checks.—“(1) Definition.—In this subsection—“(A) the term “remotely created check” means a check, including a paper or electronic check and any other payment order that the Bureau, by rule, determines is appropriately covered under this subsection, that—“(i) is not created by the financial institution that holds the customer account from which the check is to be paid; and “(ii) does not bear a signature applied, or purported to be applied, by the person from whose account the check is to be paid; and “(B) the term “Federal consumer financial law” has the meaning given the term in section 1002 of the Consumer Financial Protection Act of 2010 (12 U.S.C.).</li> <li>• 1693(i) is amended—“(1) in subsection (a)(2)(A), in the matter preceding clause (i), by striking “The term” and inserting “Subject to subsection (d)(1), the term”; “(2) by redesignating subsection (d) as subsection (e); and “(3) by inserting after subsection (c) the following: “(d) Additional Fees Prohibited.—“(1) Definition.—In this subsection, the term “prepaid account” has the meaning given the term by rule of the Bureau.</li> </ul> |
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Fig 4. Comparison between gold summary and HiLegalSum generated summary for “Stopping Abuse and Fraud in Electronic Lending Act of 2018”

In Fig. 4. presents a side-by-side comparison between the Gold Summary (the human-written reference summary) and the HiLegalSum generated summary for the legislative document titled “Stopping Abuse and Fraud in Electronic Lending Act of 2018.” The Gold Summary provides a concise overview highlighting the bill’s purpose, key amendments, and consumer protection measures. In contrast, the HiLegalSum summary automatically extracts and reformulates detailed clauses, definitions, and amendments from the original legal text. The comparison demonstrates HiLegalSum capability to retain essential legislative details and structure while significantly reducing document length, effectively summarizing complex legal language into a more manageable format.

## VI. ABLATION STUDY

To evaluate the contribution of each component in HiLegalSum, an ablation study was conducted by incrementally adding modules and observing their impact on summarization performance. The evaluation was performed on a 500-sample subset of the BillSum dataset using ROUGE-L and BERTScore (F1) as the primary metrics.

TABLE II. ABLATION STUDY

| Variant | Description                   | Rouge-L | BERTScore(F1) |
|---------|-------------------------------|---------|---------------|
| 0       | Base TF-IDF                   | 0.1921  | 0.8241        |
| 1       | +Position                     | 0.2017  | 0.8289        |
| 2       | +Legal Keywords               | 0.2089  | 0.8332        |
| 3       | +Graph Centrality(HiLegalSum) | 0.2243  | 0.8329        |

Fig 5. Impact of Position, Legal Keywords, and Graph Centrality on HiLegalSum Performance

The Ablation Study table shown in Fig 5 presents an analysis of how different components contribute to the overall performance of the HiLegalSum model. Each variant incrementally adds a new feature to the base system to assess its impact on summarization quality, measured using ROUGE-L and BERTScore (F1) metrics. The Base TF-IDF model serves as the foundation, achieving a ROUGE-L score of 0.1921 and a BERTScore of 0.8241. When positional weighting is introduced, the scores slightly improve, indicating that the sentence position within a document provides useful contextual importance. Adding legal keyword enhancement further boosts both metrics, showing that incorporating domain-specific terminology helps the model focus on legally significant sentences. Finally, the inclusion of graph centrality the core idea behind HiLegalSum results in the highest ROUGE-L score (0.2243), reflecting better structural understanding and sentence relevance.

Overall, this table demonstrates that each component incrementally enhances the summarization quality, with graph centrality and legal keyword integration contributing most significantly to HiLegalSum’s superior performance.

### A. Experimental Setup

The ablation analysis began with a Base TF-IDF model, which ranked sentences solely based on their term frequency-inverse document frequency scores. Subsequently, additional components were integrated one by one to examine their individual effect on performance.

### B. Results and Observation

**Base TF-IDF:** This baseline served as the foundation for sentence importance estimation. It captured word-level relevance but lacked contextual understanding and sentence inter-dependence, resulting in the lowest ROUGE-L (0.1921) and BERTScore (0.8241).

+ **Position:** Introducing positional weighting slightly improved performance (ROUGE-L = 0.2017, BERTScore = 0.8289), confirming that sentences appearing earlier in legal documents often carry crucial contextual or summary-worthy information.

+ **Legal Keywords:** Incorporating legal keyword awareness provided a notable boost (ROUGE-L = 0.2089, BERTScore = 0.8332). This demonstrates the effectiveness of domain-specific lexical cues in emphasizing legally informative sentences that general summarizers may overlook.

+ **Graph Centrality (HiLegalSum):** Finally, integrating graph-based semantic centrality significantly enhanced ROUGE-L to 0.2243, showing that sentence inter-connectivity and central importance play a vital role in capturing the global context of legal documents. The

BERTScore remained stable (0.8329), indicating that semantic quality was preserved while improving lexical coherence.

### C. Discussion

The ablation trend clearly demonstrates the incremental value of each module. Legal keyword awareness and graph-based centrality contribute to the most substantial improvements. Validating the hypothesis that combining semantic centrality with legal-aware weighting yields more contextually faithful summaries.

## VII. DISCUSSION

The results affirm that integrating semantic embeddings with legal-specific heuristics improves extractive summarization quality for legal texts. While graph-based centrality alone (LexRank/TextRank) identifies salient sentences, it lacks contextual and legal awareness. [6]

HiLegalSum bridges this gap by weighting semantic, positional, and legal signals simultaneously. The modest BERTScore gains (+0.01) indicate that while semantic overlap improved, there is room for stylistic and fluency enhancement, possibly via fine-tuned generation models. Overall, HiLegalSum achieves a well-balanced trade-off between informativeness and diversity — crucial in the legal domain where redundancy can obscure factual interpretation.

## VIII. LIMITATIONS

Although HiLegalSum demonstrates strong performance in legal text summarization, a few practical constraints remain: **Extractive Nature:** As an extractive summarizer, HiLegalSum selects sentences directly from the source text rather than generating new phrases. This limits its ability to paraphrase, compress, or restructure sentences naturally.

**Keyword Dependency:** The inclusion of legal keyword weighting enhances domain relevance but may sometimes introduce redundancy. Overemphasis on recurring legal terms (e.g., “section,” “clause,” “judgment”) could bias the system toward repetitive content instead of capturing novel or contextually diverse information.

**Scalability:** To efficiently process lengthy documents, HiLegalSum operates using sentence-based chunking (up to 80 sentences per segment). While this approach ensures manageable computation, it may slightly affect coherence in extremely long documents without hierarchical optimization.

## IX. CONCLUSION AND FUTURE WORK

In conclusion, text summarization plays a crucial role in simplifying and condensing lengthy and complex documents, especially in the legal domain where precision and contextual understanding are essential. Traditional methods often struggle with the dense, technical language of legal texts, leading to incomplete or less coherent summaries. However, our model HiLegalSum overcome these challenges by effectively combining relevance scoring, contextual understanding, and redundancy reduction to produce accurate and meaningful summaries. Future directions can explore domain adaptation, multi-document summarization, abstractive extension. HiLegalSum thus provides a robust,

interpretable foundation for next-generation legal summarization systems that balance domain relevance, coherence and transparency.

## X. ACKNOWLEDGEMENT

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